

Starter Panel for 60–100 HP Motor

ELECTRICAL PANEL STARTER PANEL FROM S&S GROUPS



Description About STARTER PANEL

- Design, manufacture, testing, supply, installation and commissioning of **Motor Starter Control Panel** suitable for operating **60 HP to 100 HP induction motor**, complete with all switchgear, control wiring, protection devices, busbar system, enclosure and accessories as per relevant **IS / IEC standards**.
- The panel shall be suitable for **415V $\pm 10\%$, 3 Phase, 4 Wire, 50Hz AC supply** and designed for continuous industrial operation.

- Panel Construction
- Incomer Supply
- Busbar System
- Protection System
- Motor Starter System
- Control Components

Left Sidview

Rear View

- Internal Wiring & Testing
- Motor Capacity
- Accessories & Applicable Standards
- Supply Scope & Warranty

Gland Plate

Right Side



1. PANEL CONSTRUCTION

❑ Enclosure

- Fabricated from **CRCA sheet steel**
- Sheet thickness:
 - ✓ 1.6 mm for structure
 - ✓ 1.6 mm for doors and covers
- Floor mounted, **dust and vermin proof**
- Protection class **IP54**
- Powder coated with **7 tank anti-corrosion treatment**
- Shade **RAL 7032 / Siemens Grey**

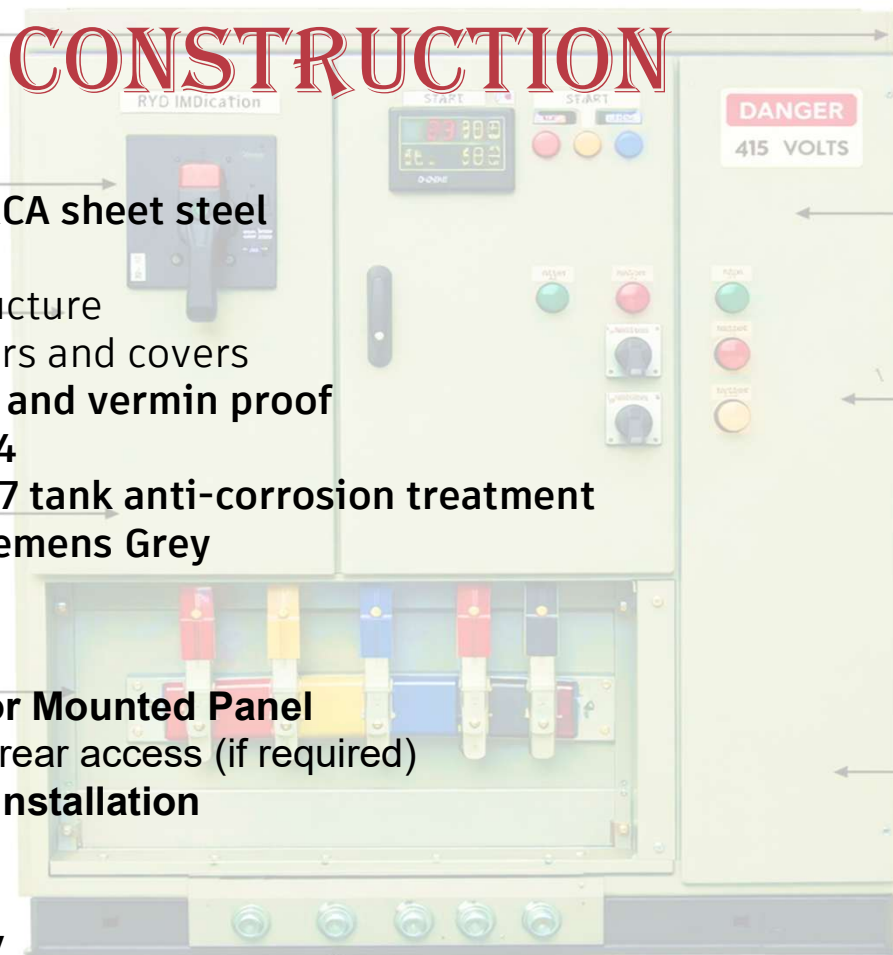
❑ PANEL TYPE

- **Free Standing Floor Mounted Panel**
- Front operated with rear access (if required)
- Suitable for **indoor installation**

❑ CABLE ENTRY

- **Bottom cable entry**
- Removable gland plates

RAL 7032 Siemens Grey



IP54
Protection

Danger
Notice Plate

Removable
Gland Plates

Bottom
Cable Entry

Free Standing
Floor Mounted

Removable Gland Plates



Busbar System – Starter Panel for 60–100 HP Motor

2. BUSBAR SYSTEM

As per Indian Standard (IS 8623, IS 4237, IEC 61439)

□ Main Busbar

- **Material:** Aluminum
- **Type:** Electrolytic grade aluminum busbar
- **Current Rating:** Suitable for motor load and panel rating
- **Busbar insulation:** Heat shrink sleeve / PVC insulation
- **Mounting:** Supported on **SMC / DMC insulated busbar supports**
- **Color coding:**
 - ✓ R – Red
 - ✓ Y – Yellow
 - ✓ B – Blue
 - ✓ N – Black

□ Earth Busbar : -

- **GI / Copper earth strip** provided throughout panel
- Earthing terminals provided for all doors and components

SMC / DMC
Insulated Supports

Aluminum Busbar
(Electrolytic Grade)

Busbar Color Coding

- R – Red
- Y – Yellow
- B – Blue
- N – Black
- Earth – Green/Yellow

Earth Busbar
(GI/Copper)

Control Terminal Blocks

3. INCOMING SUPPLY

❑ INCOMER :-

- TPN MCCB
- Rating: **250A / 200A (depending motor capacity)**
- Breaking Capacity: **36 kA / 50 kA**
- Adjustable thermal & magnetic protection
- Rotary handle door interlocked

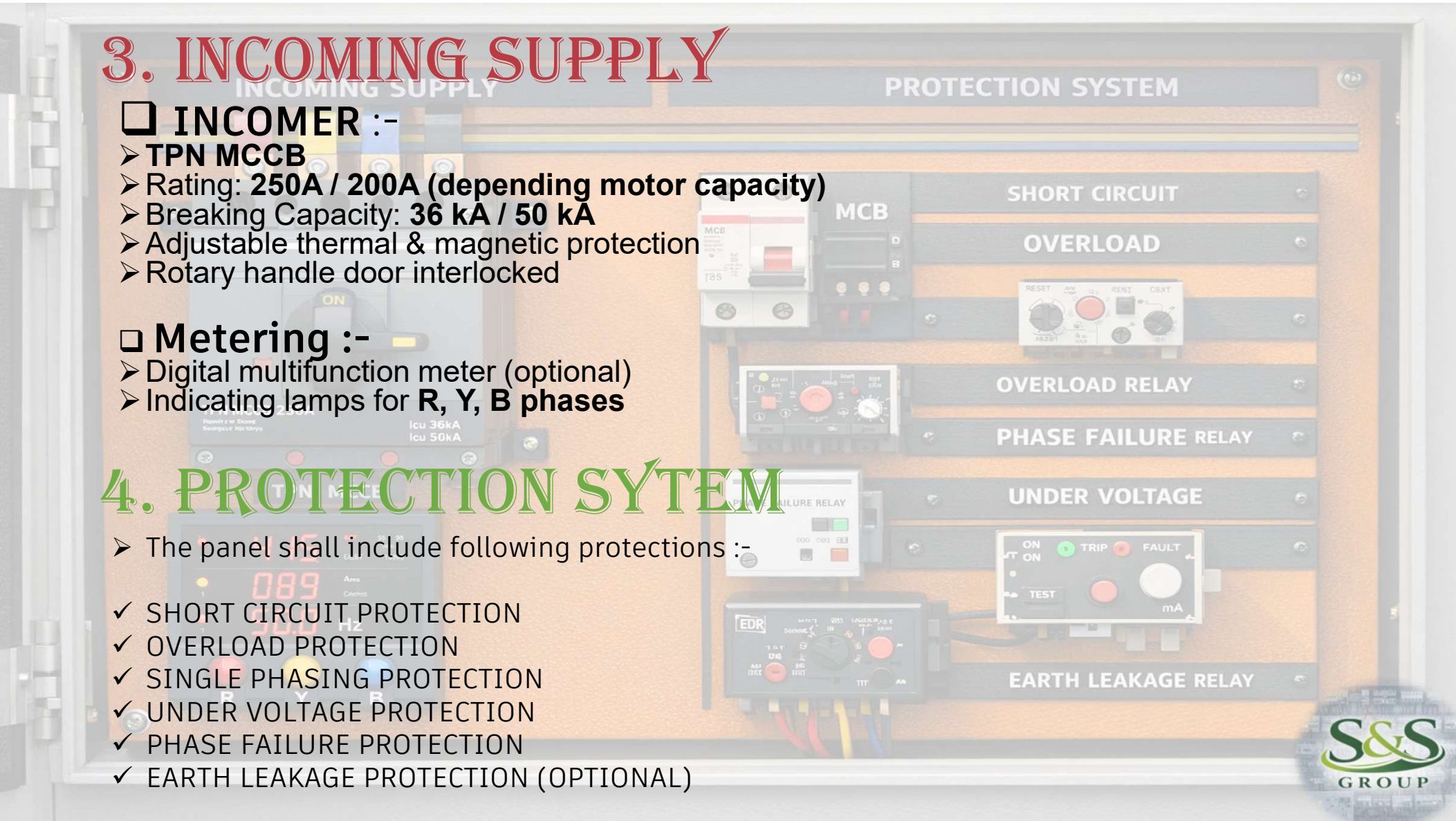
❑ Metering :-

- Digital multifunction meter (optional)
- Indicating lamps for **R, Y, B** phases

4. PROTECTION SYTEM

- The panel shall include following protections :-

- ✓ SHORT CIRCUIT PROTECTION
- ✓ OVERLOAD PROTECTION
- ✓ SINGLE PHASING PROTECTION
- ✓ UNDER VOLTAGE PROTECTION
- ✓ PHASE FAILURE PROTECTION
- ✓ EARTH LEAKAGE PROTECTION (OPTIONAL)



5. MOTOR STARTER SYSTEM

❑ STARTER TYPE SHALL BE SUITABLE FOR **60–100 HP MOTOR LOAD** :-

❖ POSSIBLE CONFIGURATIONS :-

➤ Option 1 – Star Delta Starter

✓ 3 Nos. Power Contactors

✓ Star Delta Timer

✓ Thermal Overload Relay

✓ Mechanical & electrical interlocking

➤ Option 2 – Soft Starter (optional)

✓ Soft starter suitable for **100 HP motor**

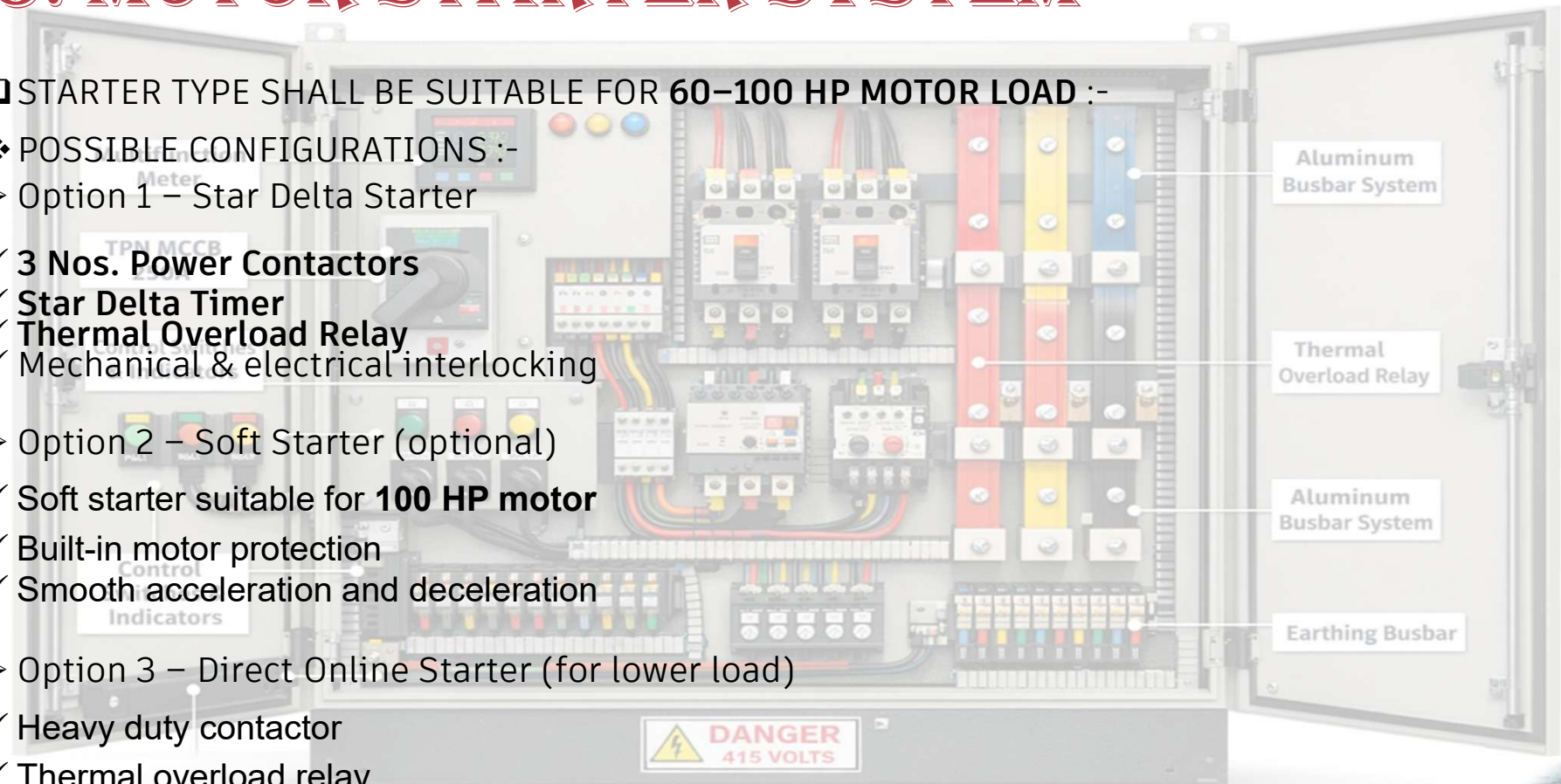
✓ Built-in motor protection

✓ Smooth acceleration and deceleration

➤ Option 3 – Direct Online Starter (for lower load)

✓ Heavy duty contactor

✓ Thermal overload relay



6. CONTROL COMPONENTS

➤ Push buttons

- ✓ START
- ✓ STOP
- ✓ RESET

➤ Indicating lamps

- ✓ Motor ON
- ✓ Trip indication
- ✓ Supply healthy

➤ Control MCB

➤ Selector switches (Auto / Manual if required)

7. INTERNAL WIRING & TESTING

❑ INTERNAL WIRING :-

- Control wiring with **FRLS copper wires**
- Ferruled wiring with identification numbers
- PVC channels for cable routing
- Terminal blocks for external connections

❑ TESTING :-

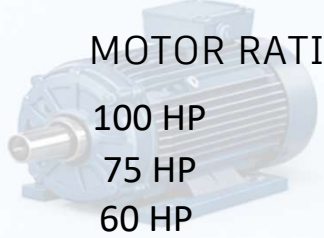
- Panel shall be tested for :-
 - ✓ Insulation resistance test
 - ✓ High voltage test
 - ✓ Functional test
 - ✓ Routine factory testing
- AS PER **IS 8623 / IEC 61439**

TESTING

8. MOTOR CAPACITY

➤ SUITABLE FOR :-

MOTOR RATING



100 HP
75 HP
60 HP

Motor Rating	Current Approx.
60 HP	~ 80–90 Amp
75 HP	~ 100 Amp
100 HP	~ 125–140 Amp

9. ACCESSORIES & APPLICABLE STANDARD

❑ ACCESSORIES :-

• Insulation Resistance Test

➤ PANEL SHALL INCLUDE :-

• High Voltage Test

- ✓ DANGER NOTICE PLATE
- ✓ NAME PLATE
- ✓ PANEL INTERNAL WIRING DIAGRAM
- ✓ CABLE GLAND PLATES
- ✓ EARTHING STUDS
- ✓ RUBBER GASKETS
- ✓ PANEL LIFTING HOOKS

DANGER
CURRENT APPROX.

~125–140 Amp
~100 Amp
~80–90 Amp

Name Plate

Wiring Diagram

Cable Glands

❑ Applicable Standards :-

➤ Manufactured as per :-

• IS 8623
• IEC 61439
• IS 4237
• IS 375
• IS 1248

- ✓ IS 8623
- ✓ IEC 61439
- ✓ IS 4237
- ✓ IS 375
- ✓ IS 1248

S&S
GROUP

SUPPLY SCOPE & WARRANTY

10. SUPPLY SCOPE & WARRANTY

SUPPLY SCOPE

➤ The scope shall include :-

- 1 Design & Engineering
 - ✓ Design And Engineering Of Starter Panel
- 2 Panel Fabrication
 - ✓ Fabrication Of Enclosure
- 3 Switchgear & Busbar Supply
 - ✓ Supply Of Switchgear And Busbar System
 - ✓ Internal Wiring And Assembly
- 4 Factory Testing
 - ✓ Factory Testing
 - ✓ Delivery At Site
- 5 Installation And Commissioning (If Required)

6 Delivery & Installation

- **WARRANTY :-**
- Minimum **12 months warranty** from date of commissioning.



WARRANTY



12 Months Warranty



On-Site Support



Free Repair / Replacement



As per Indian Standards (IS / IEC)

